

KOUTIALA, Mali

WMO: 612930

Lat: 12.383N

Lon: 5.467W

Elev: 1204

StdP: 14.07

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.5	63.6	22.6	17.8	85.1	26.1	21.0	84.4	14.5	81.8	13.3	82.4	2.5	0	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.5	104.1	70.9	102.7	71.2	101.3	70.9	80.1	91.6	79.2	90.6	78.7	89.9	6.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	149.8	84.5	76.7	145.8	83.7	76.1	142.6	83.3	44.7	91.7	43.8	90.4	43.1	89.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.9	11.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	77.1	83.1	89.0	91.4	89.9	85.3	81.1	79.7	81.0	84.5	82.5	78.3
	DBStd	5.49	3.44	4.53	3.75	3.20	3.62	3.66	2.86	2.35	2.59	2.52	2.54	3.19
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12250	839	926	1208	1242	1237	1059	966	921	930	1071	975	876
	CDD65	6775	374	506	743	792	772	609	501	456	480	606	525	411
	CDH74	85943	4013	6553	11271	12350	11442	7935	5201	4044	4681	7358	6531	4564
	CDH80	47204	1925	3773	7213	8112	7154	4153	2039	1284	1837	3747	3642	2325
Wind	WSAvg	5.4	5.0	5.3	5.0	5.4	7.2	7.1	6.4	5.5	4.5	4.6	3.8	4.6
Precipitation	PrecAvg	35.0	0.0	0.0	0.2	0.8	2.7	4.9	8.3	9.6	6.3	2.2	0.2	0.0
	PrecMax	52.9	0.4	1.1	0.7	3.3	6.2	9.6	13.5	20.7	13.7	13.5	1.7	1.7
	PrecMin	20.3	0.0	0.0	0.0	0.0	0.4	1.0	3.3	0.0	1.7	0.3	0.0	0.0
	PrecStd	6.4	0.1	0.2	0.2	0.8	1.2	1.8	2.4	3.4	2.6	2.3	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.4	103.5	105.1	105.9	105.1	101.4	96.0	91.1	93.9	98.6	98.7	96.9
		MCWB	63.9	68.6	68.8	70.7	72.4	76.7	76.8	77.8	78.2	71.9	68.4	65.7
	2%	DB	94.3	100.7	103.4	104.3	103.2	98.5	92.7	89.5	91.7	97.0	97.1	95.0
		MCWB	63.1	66.6	68.5	71.3	74.3	76.7	77.2	77.9	78.1	73.1	68.0	64.7
	5%	DB	92.0	98.8	102.1	102.8	101.3	96.3	90.6	88.0	90.2	95.4	95.7	93.3
		MCWB	62.0	65.7	68.3	71.7	75.5	76.9	76.9	77.5	77.9	74.5	67.7	64.1
	10%	DB	89.9	96.6	100.5	101.3	99.3	94.1	88.7	86.4	88.6	93.8	94.4	91.2
		MCWB	61.3	64.4	67.2	71.8	75.9	76.9	76.8	77.0	77.5	75.2	67.4	63.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.8	74.8	77.2	80.0	80.8	80.8	79.1	80.6	80.6	81.1	77.7	71.1
		MCDB	87.2	95.9	97.1	97.5	96.9	93.2	89.0	87.1	89.3	91.3	91.2	91.3
	2%	WB	67.1	72.1	75.3	78.1	79.5	79.5	78.3	79.0	79.5	79.2	75.3	68.6
		MCDB	84.6	92.5	95.2	95.0	94.9	91.7	87.7	86.4	88.3	90.0	88.9	88.8
	5%	WB	64.3	69.6	73.8	77.0	78.7	78.7	77.8	78.2	78.8	78.2	73.7	66.3
		MCDB	87.1	90.7	94.1	94.3	93.5	91.0	87.0	85.4	87.6	89.1	87.0	88.6
	10%	WB	62.5	66.6	72.3	76.0	77.9	78.0	77.3	77.4	78.0	77.4	71.7	64.5
		MCDB	86.5	92.3	92.3	93.5	92.7	90.0	86.1	84.3	86.4	88.1	85.4	87.9
Mean Daily Temperature Range	5% DB	MDBR	25.4	25.2	23.4	19.5	18.4	16.5	14.2	12.7	14.9	19.1	25.8	26.8
		MCDBR	29.1	27.8	24.8	21.8	20.1	19.0	16.5	14.9	17.3	21.4	26.9	29.3
		MCWBR	9.5	9.3	8.7	7.8	5.8	5.2	5.2	5.2	6.2	6.2	7.7	8.8
	5% WB	MCDBR	26.7	22.3	19.7	18.8	18.6	16.8	15.2	14.0	16.3	18.3	21.6	27.7
		MCWBR	10.8	9.5	8.7	6.6	5.7	5.9	5.7	5.7	6.5	6.4	7.9	10.0
Clear-Sky Solar Irradiance	taub		0.524	0.582	0.715	0.710	0.674	0.641	0.571	0.537	0.536	0.570	0.527	0.500
	taud		1.880	1.743	1.517	1.605	1.662	1.757	1.943	2.036	2.022	1.909	1.910	1.942
	Ebn at Noon		246	238	211	211	215	219	236	248	250	238	244	249
	Edh at Noon		62	74	95	86	80	72	60	55	56	62	60	57
All-Sky Solar Radiation	RadAvg		1775	1900	1998	2000	1969	1864	1781	1702	1795	1881	1803	1733
	RadStd		99	94	95	96	55	52	47	72	87	47	54	63

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page